

Graphing the Parabola

A Structural Approach to Algebra

$$y = 5x^2 - 20x + 15$$

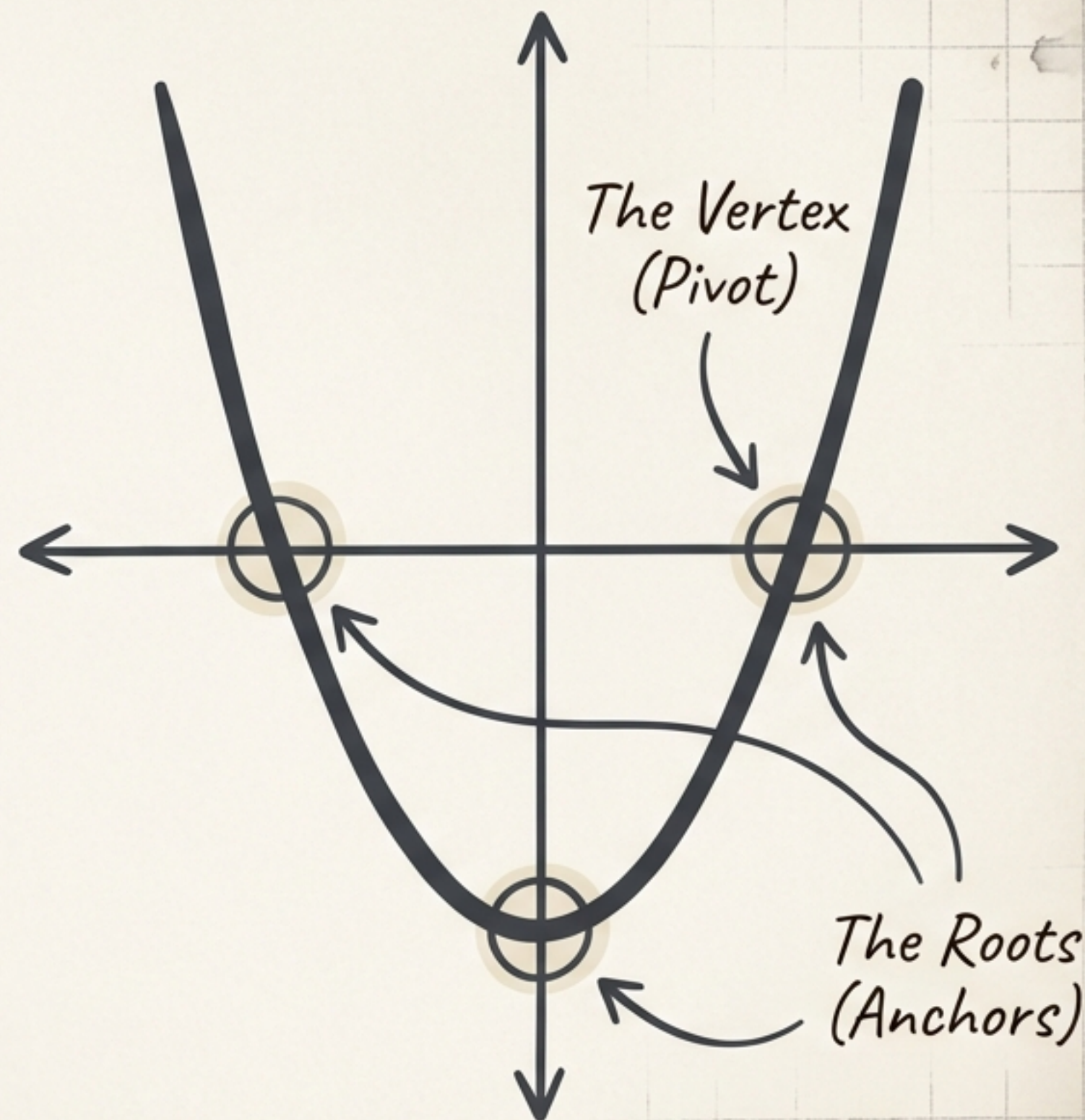
A visual guide to translating an algebraic string into a geometric shape.

The Strategy

Defining Features vs. Random Points

We could solve this by brute force: picking random values for x , calculating y , and hoping the shape emerges. However, a smarter approach is to identify the graph's structural anchors:

1. **The Roots:** Where the graph touches the ground.
2. **The Vertex:** The precise turning point of the curve.



Phase 1: Finding the Anchors

To find the roots, we need to know where the graph intersects the x-axis. Geometrically, intersecting the x-axis means the **height (y)** is zero.

$$0 = 5x^2 - 20x + 15$$



*Height is set
to zero*

Simplification

$$0 = \boxed{5x^2} - 20x + 15$$

$$\frac{0}{5} = \frac{5x^2}{5} - \frac{20x}{5} + \frac{15}{5}$$

$$0 = x^2 - 4x + 3$$

Dividing by the coefficient (5) simplifies the factoring process without changing the roots.

The Sum & Product Logic

$$x^2 - 4x + 3 = 0$$

Target
Product

$$+3$$

$$(-3) \times (-1) = 3$$

Target
Sum

$$-4$$

$$(-3) + (-1) = -4$$

$$(x - 3)(x - 1) = 0$$

Solving for X

The Zero Product Property

$$(x - 3)(x - 1) = 0$$

If this term
is zero...

$$x - 3 = 0$$

$$x = 3$$

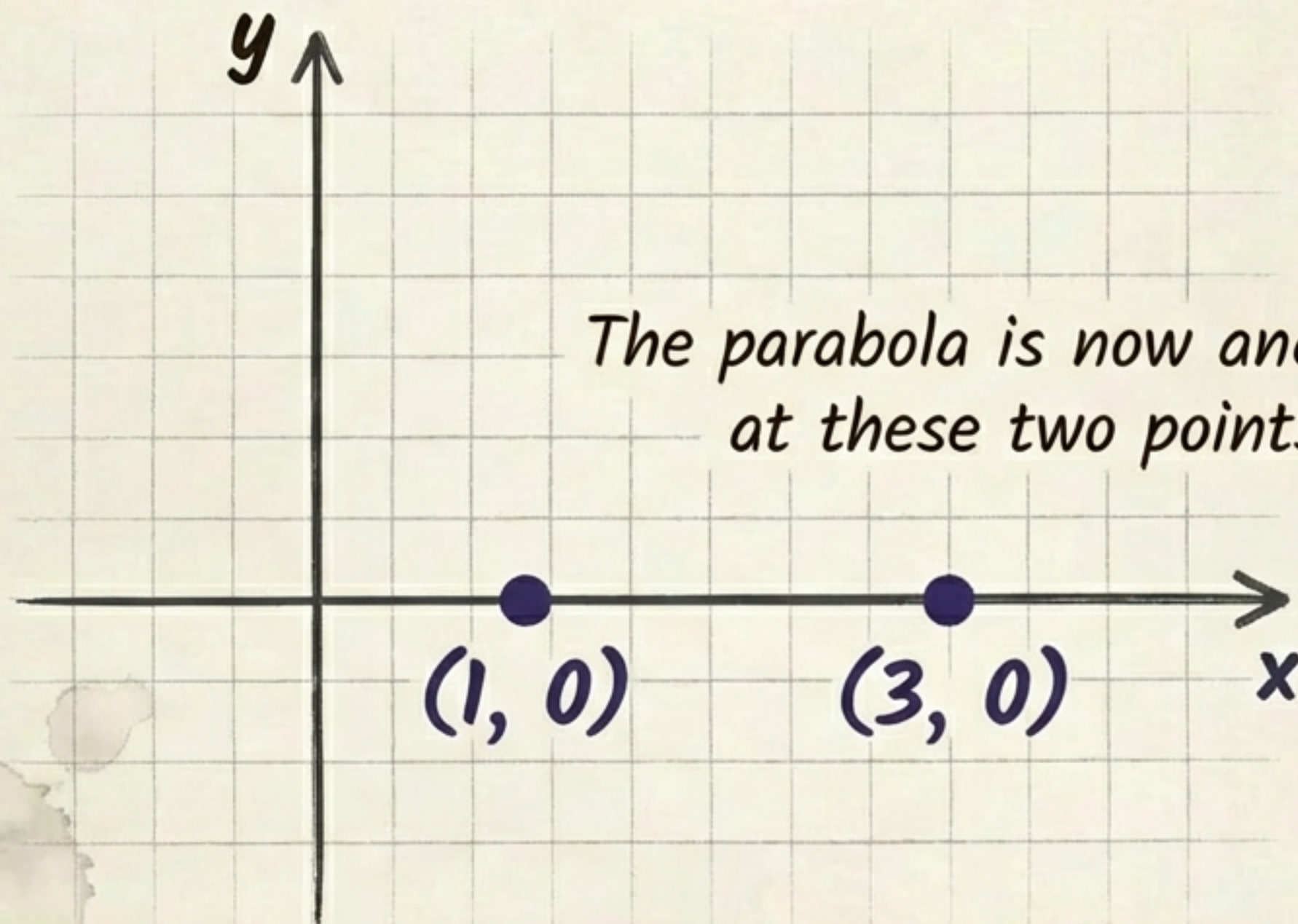
Or if this term
is zero...

$$x - 1 = 0$$

$$x = 1$$

We have established two distinct locations where the graph crosses the axis.

Visual Checkpoint: The Roots



Phase 2: The Symmetry Insight

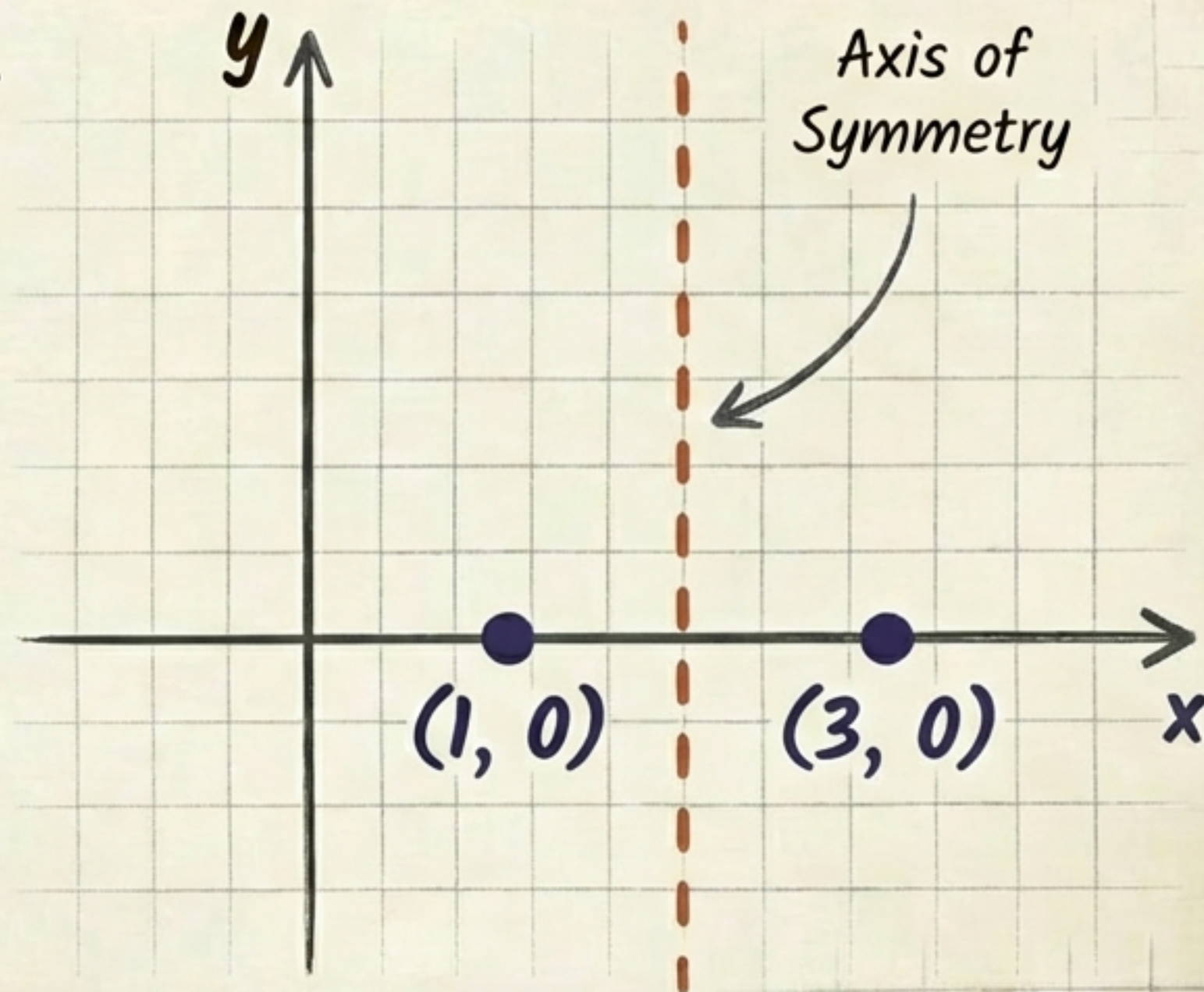
Parabolas are perfectly symmetrical. The turning point (Vertex) must lie exactly halfway between the roots.

Root A: $x = 1$

Root B: $x = 3$

Midpoint: $(1 + 3) / 2 = 2$

Vertex x-coordinate = 2



Finding the Y-Coordinate

Substitution

$$y = 5x^2 - 20x + 15$$

$$y = 5(2)^2 - 20(2) + 15$$

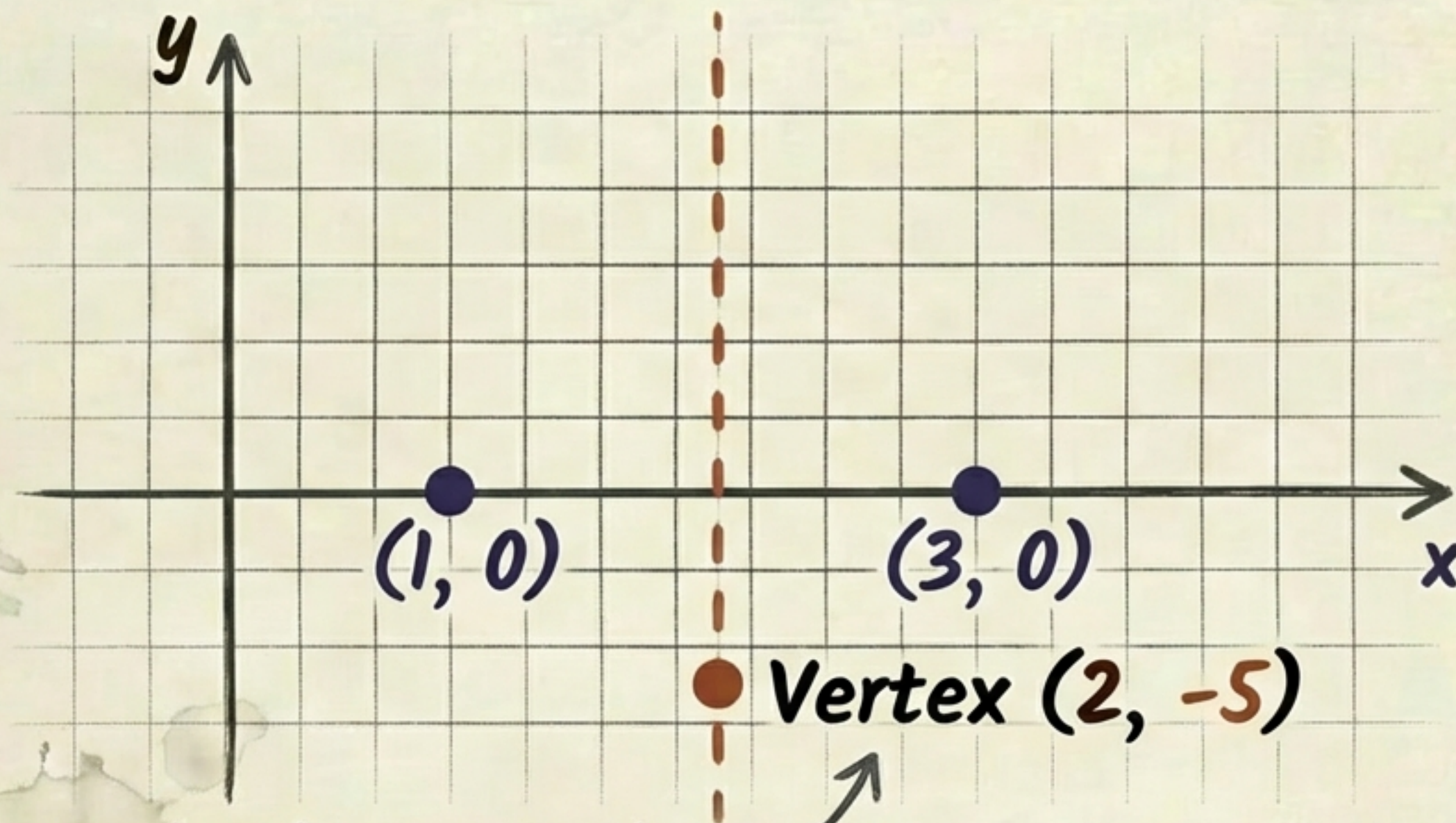
$$y = 5(4) - 40 + 15$$

$$y = 20 - 40 + 15$$

$$y = -5$$

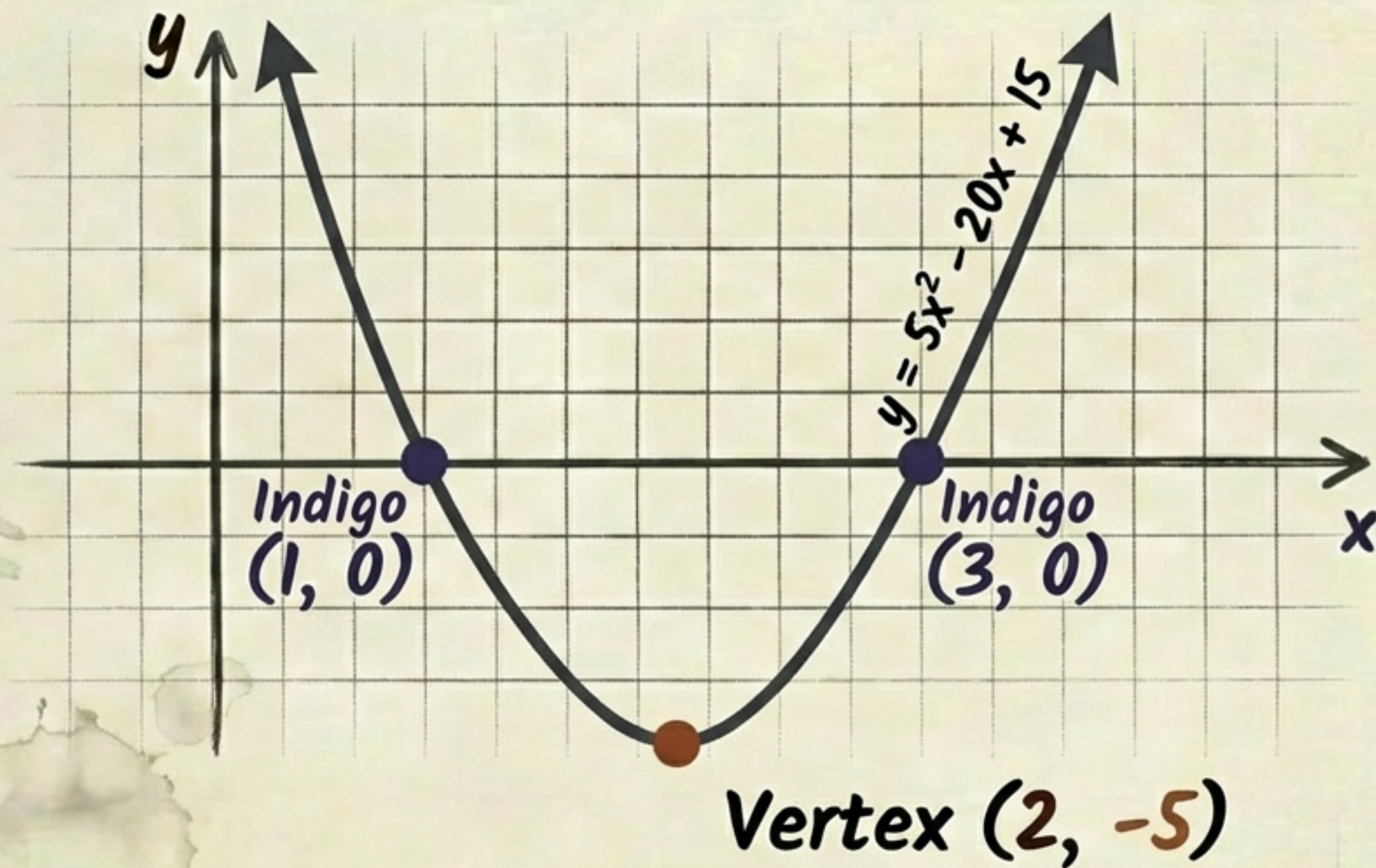
$$\text{Vertex} = (2, -5)$$

Visual Checkpoint: The Vertex



We now have the three critical points required to define the curve.

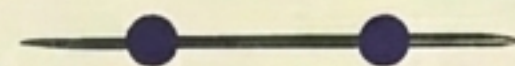
The Final Graph



Recipe: The Roots & Vertex Method

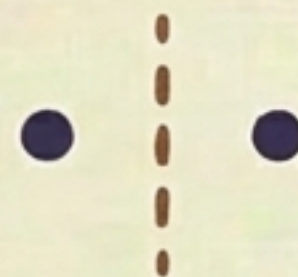
1. Find the Roots

Set $y = 0$. Factor the equation to find x -intercepts.



2. Locate the Axis

Find the midpoint between the two roots.



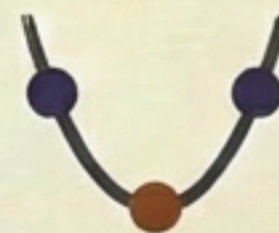
3. Calculate Vertex Height

Substitute the midpoint x -value back into the original equation to find y .

$$f(x) \rightarrow y$$

4. Connect the Points

Draw the curve through the roots and vertex.



This method reveals the structure of the parabola without guessing.